

Poissonian image deconvolution with analysis sparsity priors

Houzhang Fang
Luxin Yan

Huazhong University of Science and Technology
Institute for Pattern Recognition and Artificial Intelligence, Science and Technology on Multi-spectral
Information Processing Laboratory
Wuhan, Hubei 430074, China
E-mail: yanluxin@gmail.com

Abstract. Deconvolving Poissonian image has been a significant subject in various application areas such as astronomical, microscopic, and medical imaging. In this paper, a regularization-based approach is proposed to solve Poissonian image deconvolution by minimizing the regularization energy functional, which is composed of the generalized Kullback-Leibler divergence as the data-fidelity term and sparsity prior constraints as the regularization term, and a non-negativity constraint. We consider two sparsity prior constraints which include framelet-based analysis prior and combination of framelet and total variation analysis priors. Furthermore, we show that the resulting minimization problems can be efficiently solved by the split Bregman method. The comparative experimental results including quantitative and qualitative analysis manifest that our algorithm can effectively remove blur, suppress noise, and reduce artifacts. © 2013 SPIE and IS&T [DOI: [10.1117/1.JEI.22.2.023033](https://doi.org/10.1117/1.JEI.22.2.023033)]

1 Introduction

Image deconvolution plays an important role in many applied fields of the signal and image processing.^{1–5} In these applications, the underlying images u are degraded by blur, because of the imaging system, and corrupted by Poisson noise process $\mathcal{P}$. Therefore, the image formation process can be modeled as

$$g = \mathcal{P}(Hu), \quad (1)$$

where $g: \Omega \rightarrow R$ is the observed image. Here, $\Omega \in \mathbb{R}^2$ is a bounded rectangular domain and H contains the information on the point spread function (PSF) determining the blur on the observed image.

Poissonian image deconvolution is an ill-posed inverse problem where small perturbations in the input data, due to noise, can cause large variations in the solution. A standard approach for solving an inverse problem is to express the desirable solution as the minimizer of a regularization energy functional. The regularization term is introduced in order to enforce stability as well as incorporate prior knowledge about the solution. Under Poisson noise, the regularization framework can be expressed as

$$\min_{u \geq 0} \int_{\Omega} [Hu - g \log(Hu)] dx + \lambda R(u), \quad (2)$$

where $u \geq 0$ guarantees that the restored image does not contain negative intensities. $R(u)$ is the regularization constraint about the underlying solution u . $\lambda > 0$ is a regularization parameter. Let $v = Hu$, then

$$\begin{aligned} D(g, v) &= \int_{\Omega} (g \log \frac{g}{v} - g + v) dx \\ &= \int_{\Omega} (v - g \log v + g \log g - g) dx \end{aligned}$$

is called the generalized Kullback-Leibler (KL) divergence. If it is finite, it coincides with the Bregman distance⁶ of the Boltzmann-Shannon entropy. Hence, we have $D(g, v) \geq 0$. Moreover, the observed image can be seen as the set of samples of Poisson variables, thus, $g \geq 0$. We set $\ln(0) = -\infty$ and $0 \ln(0) = 0$.

Existing image priors usually model images either in an analysis way or a synthesis way⁷ and, in both kinds of priors, sparsity is widely used. An analysis based prior seeks a signal or image whose forward (analysis) transform coefficients are sparse, such as gradients,⁴ while synthesis based priors decompose each image into a sparse linear combination of atoms in a dictionary and directly model the probability distribution of combination coefficients.

Over the years, various deconvolution algorithms have been developed to solve the Poissonian image deconvolution problem, such as the classical Richardson-Lucy (RL) algorithm, which can be found in Ref. 8. However, the RL algorithm amplifies noise after a few iterations which can be avoided by introducing regularization. In Ref. 2, the authors proposed to combine the RL algorithm with a regularization constraint, based on total variation (TV), which preserves the object edges well. In Ref. 9, Sawatzky proposed an EM-TV algorithm to solve the minimization problem [Eq. (2)]. Chen et al.¹⁰ proposed a TV-KL model with a spatially dependent regularization parameter which adopted the TV model $R(u) = \int_{\Omega} |\nabla u|$ in Eq. (2) to regularize the images. Yan et al.⁵ proposed the improved RL blind image deconvolution algorithm by employing spatially adaptive TV based on the edge indicator operator to reduce the noise in flat regions as

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well as preserve the edge and detailed information. Several authors also proposed the wavelet based regularization.^{11,12}

The framelet based image restoration has been developed very fast in the last decade for its robustness and stableness. One of the advantages of using framelet systems is that they provide reasonably good sparse approximations to piecewise smooth functions, which form a large class of functions in which the images belong. This stimulates the extensive research on framelet based image restoration.^{13–17} Since the framelet systems are redundant systems, the representation of image u in the framelet domain is not unique. Therefore, there are three formulations for the sparse approximation of the underlying images which include analysis based approach, synthesis based approach, and balanced approach. The balanced approach can be viewed as a way to balance the analysis and synthesis based approaches.^{14,17} Dupé et al.³ proposed to use the Anscombe variance stabilizing transform which allows using a nonlinear observation model with additive noise to replace a linear observation model with Poisson noise. Figueiredo et al.¹³ proposed an approach to deconvolve Poissonian images by using the augmented Lagrangian method with the alternating method of multipliers method. They considered the regularizers which are based on either an analysis prior or a synthesis prior. It is noted, that the framelet based approaches solve image restoration problems in digital domain directly and many associated efficient algorithms which utilize the multilevel nature of wavelet frame transforms are developed to achieve better sparse approximation of the underlying solutions.^{17,18}

The standard criterion for the deconvolution problem of Poissonian images can be formulated as an optimization problem of a convex functional which combines a nonlinear data-fidelity term characterizing the noise properties and a convex nonsmooth regularization term and a constraint term enforcing the solution to be non-negative. Then, Eq. (2) can be further modified as

$$\min_u \int_{\Omega} Hu - g \log(Hu) dx + \lambda R(u) + l_{u \geq 0}, \quad (3)$$

where the role of $l_{u \geq 0}$ is to impose the non-negative constraint on the estimation image u . When dealing with images, we will consider our problems in a discrete setting and employ the usual vector notation obtained by stacking the images pixels columnwise into vectors in $\mathbb{R}^N$ in lexicographic order. The discrete version of problem [Eq. (3)] is given by

$$\min_{u \in \mathbb{R}^N} \langle 1, Hu - g \log Hu \rangle + \lambda R(u) + l_{u \geq 0}, \quad (4)$$

where $u \in \mathbb{R}^N$ is the underlying image to be recovered, $H \in \mathbb{R}^{N \times N}$ is a blur matrix, “1” stands for an N -size vector whose components are all equal to one. In the framework of Eq. (4), Setzer et al.⁴ exploited TV as the regularization and proposed an efficient algorithm based on split Bregman techniques, which took the Poisson noise statistics into account. Figueiredo et al.¹³ also addressed their deconvolution problem in the form of Eq. (4). Both of them reported the impressive effect.

However, as is well known, the main drawback of TV regularization is that it can not preserve the fine structures, textures, and details very well on the regions of complicated structures due to the so-called staircasing effect [As shown in

Fig. 1(b)]. Thus, some important image features, such as edge and textures, will be blurred in the deconvolution process. The goal of this paper is to develop efficient algorithms for solving Poissonian image deconvolution problems. In the present work, we tackle such ill-posed inverse problem in the regularization framework of Eq. (4). In order to overcome the ill-posedness of the problem, proper regularization terms on the underlying image have to be enforced in minimization model [Eq. (4)]. The basic idea of our approach is to utilize the sparsity prior of the images to regularize the images. Recently, the impressive effects have been achieved by employing sparsity prior of images under the framelet system in many image restoration problems.^{14–17} Since the framelet transform has the ability of multiple-resolution analysis in nature, different framelet masks reflect different orders of difference operators which can adaptively capture multi-scale edge structures in the latent image. Therefore, the framelet has potential to preserve more details and texture information of the image compared with total variation. Thus, the ℓ_1 -norm of framelet coefficients of the latent images is utilized as one of the sparse regularization constraint terms in the Poissonian image deconvolution. In fact, the extensively used TV based regularization can also be regarded as a sparsity analysis prior constraint on image gradients because it penalizes ℓ_1 -norm of the gradient transform coefficients of the image. Inspired by the edge-preserving advantage of TV and the capability of sparsely approximating piecewise smooth images of the framelet based method, we propose to combine the framelet based sparsity constraint with TV for Poissonian image deconvolution because this combination will significantly favor reducing artifacts effect yielded by only using TV regularization.

The nonquadratic and nonseparable Poisson log-likelihood and the convex nonsmooth regularizer in Eq. (4)



Fig. 1 Poissonian image deconvolution using the method of Ref. 4. (a) Lena image is blurred with Gaussian kernel with the size of 15×15 and the standard deviation equals to 1.8 and then corrupted by Poisson noise. (b) Restored image by the method of Ref. 4.

present challenging computational difficulty to our image deconvolution algorithms. For the large-scale problems of interest to us, the computation of the minimizer typically requires the use of an optimization scheme. Recently, split Bregman method¹⁹ was proposed by Goldstein and Osher and has many successful applications.^{14,16,19} It can efficiently solve general ℓ_1 -regularized optimization problems with multiple ℓ_1 -regularized terms. Furthermore, another advantage of the split Bregman method is that it has a relatively small memory footprint and is easy to code by users.^{14,19} These good properties are significant for large scale problems. We will see that the ℓ_1 related minimization problems, resulting from our proposed models [Eq. (4)], can be efficiently solved by split Bregman method.

It has been found, previously, that coupling regularization based on wavelets often improves image quality.^{19,20} Our framelet based coupling regularization method extends the work, by Ref. 19, to redundant wavelet tight frame system. The next section describes the formulation of the image regularization term $R(u)$ in detail.

This paper is structured as follows. Two regularization functionals are presented in Sec. 2. Section 3 provides the numerical details of the minimization procedure. In Sec. 4, we illustrate the good performance of our algorithms by numerical experiments. The conclusions are given in Sec. 5.

2 Analysis Sparsity Priors-Based Poissonian Image Deconvolution Models

In this section, we introduce our framelet based and coupling regularization models based on two sparsity analysis priors.

We, now, briefly introduce the concept of tight frames and framelets. Interesting readers could consult Ref. 21 for theories of frames and framelets.

A countable set $X \subset L_2(\mathbb{R})$ is called a tight frame of $L_2(\mathbb{R})$ if

$$f = \sum_{h \in X} \langle f, h \rangle h \quad \forall f \in L_2(\mathbb{R}),$$

where $\langle \cdot, \cdot \rangle$ is the inner product of $L_2(\mathbb{R})$. The tight frame X is called a tight wavelet frame if the elements of X is generated by dilations and translations of finitely many functions called framelets. Given a one-dimensional (1-D) framelet system for $L_2(\mathbb{R})$, the n -dimensional tight wavelet frame system for $L_2(\mathbb{R}^n)$ can be easily constructed by using tensor products of 1-D framelets. In the discrete setting, a discrete image u is an n -dimensional array. We use matrix multiplication W and W^T to stand for the discrete fast tensor product framelet decomposition and reconstruction, respectively, where W^T is obtained by the wavelet filters derived by the unitary extension principle²¹ and we have the perfect reconstruction formula, $I = W^T W$, for all $u \in \mathbb{R}^n$. We will further denote an L -level framelet decomposition of u as

$$Wu = \{W_{l,i,j}u: 1 \leq l \leq L, (i, j) \in I\},$$

where I denotes the index set of all framelet bands. More details on discrete algorithms of framelet transforms can be found in Ref. 18.

2.1 Framelet-Based Poissonian Image Deconvolution Model

Poissonian image deconvolution is an ill-posed problem and is not easy to solve without regularization. In the regularization framework of Eq. (4), the desired solution can be obtained by minimizing an objective functional, which contains the log-likelihood term and a regularization term, and a positivity constraint. Recently, a great many regularization based optimization algorithms have been proposed to solve Poissonian image deconvolution problems. Among all regularization based models for image restoration, TV based methods are widely adopted and have been proven successful and effective. However, the TV often yields the so-called staircasing effect, a phenomenon which stems from the fact that TV regularization has a tendency to produce piecewise constant solutions, which is simply because piecewise constant functions can be sparsely approximated in the gradient domain. Tight wavelet frame prior has been successfully used in the image deconvolution problems.¹⁴⁻¹⁶ The prior assumes that the underlying images have a good sparse representation under the appropriate wavelet tight frame transform W . Motivated by the strong robustness to noise of the framelet,¹⁴ we use the analysis sparsity prior on the image u under framelet system to regularize the Poissonian image deconvolution. Thus, we propose the following Poissonian image restoration model that we shall focus on in the paper:

$$\min_{u \in \mathbb{R}^N} \langle 1, Hu - g \log Hu \rangle + \lambda \|Wu\|_1 + I_{u \geq 0}, \quad (5)$$

where W is the discrete framelet transform. In our implementation, we use the B-splines framelet, with which each framelet mask can be regarded as a standard difference operator that can be understood as a discretization of the corresponding differential operator.¹⁷ One of the advantages of employing the B-splines framelet is that they favor piecewise smooth images, especially, on reducing the artifacts of the restored images. For more construction details on the piecewise linear B-spline framelet, we refer to Refs. 14 and 21.

2.2 Sparsity Priors-Based Coupling Regularization Model

The TV prior based regularization is one of the popular models and has proven to be effective in suppressing noise and preserving edge while maintaining quality of the reconstructed images. In fact, the ℓ_1 -norm of the image gradient based regularization can also be viewed as a sparsity regularization on gradient vector. The TV minimization recovers an image with a gradient vector that is as sparse as possible, therefore, the TV prior model favors the underlying estimated image with a good sparse approximate solution in the gradient domain. The key advantage of TV prior regularization is that it permits discontinuities in the computed solutions and yields a sharper edge and permits to remove the artifacts. Thus, we propose to combine TV prior with wavelet frame analysis prior. In particular, we present a non-smooth coupling regularization method for Poissonian image deconvolution:

$$\min_{u \in \mathbb{R}^N} \langle 1, Hu - g \log Hu \rangle + \lambda_1 \|\nabla u\|_1 + \lambda_2 \|Wu\|_1 + I_{u \geq 0}, \quad (6)$$

where W is the B-splines framelet transform and λ_1 and λ_2 are the regularization parameters.

The TV term enhances edges in the recovered image while the inclusion of the framelet term favors restoring the abundant details and textures. Thus, combination of TV and framelet based sparsity regularization is expected to yield to the sharper edge enhancement and artifact reduction and restore more image details information. Moreover, this method will have further advantages in its implementation efficiency as there is little additional cost to include such regularizers due to the introduction of efficient solution method called split Bregman method. In fact, the coupling of such "overcomplete" transforms has little impact on the convergence rate of the split Bregman algorithm and both transforms can be implemented efficiently. This is also observed in Ref. 19 by combining TV and Haar wavelet regularizers to deal with compressed sensing problem. Similar idea can be also found in Ref. 20.

3 Alternating Split Bregman Algorithm for the Proposed Models

3.1 Standard Alternating Split Bregman Algorithm

Here, we first give a review for the alternating split Bregman algorithm.^{14,19} In the derivation of the alternating split Bregman algorithm, the authors of Ref. 19 employed a variable splitting idea of Ref. 22. Thus, the alternating split Bregman algorithm can be viewed as a splitting method, namely, it is equivalent to the Douglas-Rachford splitting method applied to the dual problem or, equivalently, to the alternating direction method of multipliers. For more details, we refer the reader to Ref. 23. Goldstein and Osher proposed the alternating split Bregman algorithm in Ref. 19 to solve the following unconstrained minimization problem:

$$\min_{u \in \mathbb{R}^N} g(u) + \Phi(Du), \quad (7)$$

where $g: \mathbb{R}^N \rightarrow \mathbb{R} \cup \{+\infty\}$, $\Phi: \mathbb{R}^M \rightarrow \mathbb{R} \cup \{+\infty\}$ are proper, convex and lower semicontinuous functionals, and $D \in \mathbb{R}^{M \times N}$. Rather than considering Eq. (7), we will consider the problem

$$\min_{u \in \mathbb{R}^N, d \in \mathbb{R}^M} g(u) + \Phi(d), \quad \text{s.t. } d = Du. \quad (8)$$

Then, the alternating split Bregman algorithm can be exploited to efficiently solve the above constraint problem:

$$\begin{cases} u^{k+1} = \arg \min_{u \in \mathbb{R}^N} g(u) + \frac{1}{2\alpha} \|b^k + Du - d^k\|_2^2, \\ d^{k+1} = \arg \min_{d \in \mathbb{R}^M} \Phi(d) + \frac{1}{2\alpha} \|b^k + Du^{k+1} - d\|_2^2, \\ b^{k+1} = b^k + (Du^{k+1} - d^{k+1}). \end{cases} \quad (9)$$

3.2 Poissonian Image Deconvolution Based on Split Bregman Algorithm

Next, we give a detailed description of numerical algorithm for the coupling regularization model [Eq. (6)]. As to Eq. (5), the numerical algorithm can be obtained by properly modifying the above derived algorithm. The difficulties for the coupling regularization method of Eq. (6) are that the term $\|\nabla u\|_1$ and $\|Wu\|_1$ are nonsmooth and nonseparable

and the data term is nonquadratic and nonseparable. Recently, an efficient optimization method, called the split Bregman algorithm, was developed to solve the ℓ_1 norm based regularization.^{4,14,16,19} The basic idea of split Bregman iteration is to convert the unconstrained minimization problem [Eq. (6)] into a constraint one by introducing auxiliary variables $d_1 = Hu$, $d_2 = \nabla u$, $d_3 = Wu$, and $d_4 = u$, and then the Bregman iteration is employed to solve the constrained optimization problem. Interested readers can refer to Refs. 14 and 19 for more details.

In the first, we need to solve the following ℓ_1 norm based minimization:

$$\min_{u \in \mathbb{R}^N} \langle 1, Hu - g \log Hu \rangle + \lambda_1 \|\nabla u\|_1 + \lambda_2 \|Wu\|_1 + l_{u \geq 0}. \quad (10)$$

We introduce four auxiliary variables d_1 , d_2 , d_3 , and d_4 , and Eq. (10) then is equivalent to

$$\begin{aligned} \min_{u \in \mathbb{R}^N, d_1, d_2, d_3, d_4} & \langle 1, d_1 - g \log d_1 \rangle + \lambda_1 \|d_2\|_1 + \lambda_2 \|d_3\|_1 + l_{d_4 \geq 0}(d_4) \\ \text{s.t. } & d_1 = Hu, d_2 = \nabla u, d_3 = Wu, d_4 = u. \end{aligned} \quad (11)$$

Further, Eq. (11) can be converted into an unconstrained problem via the Bregman iteration:

$$\begin{aligned} \min_{u \in \mathbb{R}^N, d_1, d_2, d_3, d_4} & \langle 1, d_1 - g \log d_1 \rangle + \lambda_1 \|d_2\|_1 + \lambda_2 \|d_3\|_1 \\ & + l_{d_4 \geq 0}(d_4) + \frac{1}{2\alpha} \{ \|b_1 + Hu - d_1\|_2^2 + \|b_2 + \nabla u - d_2\|_2^2 \\ & + \|b_3 + Wu - d_3\|_2^2 \} + \frac{1}{2\beta} \|b_4 + u - d_4\|_2^2. \end{aligned} \quad (12)$$

There are certain advantages to formulate the problem because the unconstrained problem [Eq. (12)] can be solved using a simple alternating minimization scheme.^{19,22} Therefore, the minimization of Eq. (12) can be performed alternately with the following five subproblems:

$$\begin{cases} u^{k+1} = \arg \min_{u \in \mathbb{R}^N} \frac{1}{2\alpha} (\|b_1^k + Hu - d_1^k\|_2^2 + \|b_2^k + \nabla u - d_2^k\|_2^2 \\ \quad + \|b_3^k + Wu - d_3^k\|_2^2) + \frac{1}{2\beta} \|b_4^k + u - d_4^k\|_2^2, \\ d_1^{k+1} = \arg \min_{d_1} \langle 1, d_1 - g \log d_1 \rangle + \frac{1}{2\alpha} \|b_1^k + Hu^{k+1} - d_1\|_2^2, \\ d_2^{k+1} = \arg \min_{d_2} \lambda_1 \|d_2\|_1 + \frac{1}{2\alpha} \|b_2^k + \nabla u^{k+1} - d_2\|_2^2, \\ d_3^{k+1} = \arg \min_{d_3} \lambda_2 \|d_3\|_1 + \frac{1}{2\alpha} \|b_3^k + Wu^{k+1} - d_3\|_2^2, \\ d_4^{k+1} = \arg \min_{d_4} l_{d_4 \geq 0}(d_4) + \frac{1}{2\alpha} \|b_4^k + u^{k+1} - d_4\|_2^2, \\ b_1^{k+1} = b_1^k + (Hu^{k+1} - d_1^{k+1}), \\ b_2^{k+1} = b_2^k + (\nabla u^{k+1} - d_2^{k+1}), \\ b_3^{k+1} = b_3^k + (Wu^{k+1} - d_3^{k+1}), \\ b_4^{k+1} = b_4^k + (u^{k+1} - d_4^{k+1}). \end{cases} \quad (13)$$

In Eq. (13), each iteration of the u subproblem is obtained by solving the following linear system:

$$\begin{aligned} [\beta(H^T H + \nabla^T \nabla + I) + \alpha W^T W] u^{k+1} &= [\beta H^T (d_1^k - b_1^k) \\ &+ \beta \nabla^T (d_2^k - b_2^k) + \beta (d_4^k - b_4^k) + \alpha W^T (d_3^k - b_3^k)], \end{aligned} \quad (14)$$

note that $W^T W = I$. Imposing periodic boundary conditions for the image, hence, Eq. (14) can be efficiently performed in the Fourier transform domain.

The second subproblem of Eq. (13) can be solved easily as follows:

$$d_1^{k+1} = \frac{1}{2} (b_1^k + H u^{k+1} - \alpha + \sqrt{(b_1^k + H u^{k+1} - \alpha)^2 + 4\alpha g}). \quad (15)$$

We can explicitly solve the minimization problem for d_2 using generalized soft-threshold formula:^{19,22}

$$\begin{aligned} d_2^{k+1} &= \text{shrink}(\nabla u^{k+1} + b_2^k, \lambda_1 \alpha) \\ &= \max(\|\nabla u^{k+1} + b_2^k\|_2 - \lambda_1 \alpha, 0) \frac{\nabla u^{k+1} + b_2^k}{\|\nabla u^{k+1} + b_2^k\|_2}. \end{aligned} \quad (16)$$

The fourth subproblem of Eq. (13) can be solved efficiently, i.e., we can solve d_3 by soft-threshold formula as follows (see, e.g., Ref. 24):

$$d_3^{k+1} = \mathcal{T}_{\lambda_2 \beta}(W u^{k+1} + b_3^k), \quad (17)$$

where the soft-thresholding operator $\mathcal{T}_\tau(v)$ is an entry-wise operation defined by

$$\mathcal{T}_\tau(v) = \frac{v}{|v|} \max\{|v| - \tau, 0\}.$$

The fifth subproblem of Eq. (13) can be computed by the direct formula:

$$d_4^{k+1} = \begin{cases} b_4^k + u^{k+1}, & \text{if } b_4^k + u^{k+1} \geq 0 \\ 0, & \text{otherwise} \end{cases}. \quad (18)$$

Taking all of the above into account, the complete resulting algorithm for the Poissonian image deconvolution is summarized as follows:

For the above algorithm, we call it PIDSB-FATV. Numerical algorithms for framelet based algorithm can be obtained by properly modifying the above Algorithm 1, which gives Algorithm 2 (we call it PIDSB-FA).

4 Experimental Results

In this section, we present numerical results of our proposed PIDSB-FA and PIDSB-FATV algorithms. For adding Poisson noise to an original image u , we first scaled the image by $(I_{\max}/u_{\max})/10^{12}$, where $u_{\max}$ is the maximal value of u and $I_{\max}$ is a specified maximal intensity (which coincides with the Poisson noise intensity). Then we applied the MATLAB command `imnoise(I, 'Poisson')` and, afterwards, scaled it back again as presented in Ref. 4. For the measure of convergence, the “normalized step difference energy” (NSDE) was calculated for each iteration: $\text{NSDE} =$

Algorithm 1 Coupling regularization for Poissonian image deconvolution using split Bregman iteration (PIDSB-FATV).

Initialize: Choose $u^0 = g$, $d_1^0 = Hg$, $d_2^0 = \nabla g$, $d_3^0 = Wg$, $d_4^0 = g$, and $b_1^0 = b_2^0 = b_3^0 = b_4^0 = 0$. Set $k = 0$.

while stopping criterion is not satisfied **do**

$k = 0$ to n_{MaxIter}

Step 1: Solve u subproblem using Eq. (14);

Step 2: $d_1^{k+1} = \frac{1}{2}(s^k + \sqrt{(s^k)^2 + 4\alpha g})$, where $s^k = b_1^k + H u^{k+1} - \alpha$;

Step 3: $d_2^{k+1} = \text{shrink}(\nabla u^{k+1} + b_2^k, \lambda_1 \alpha)$;

Step 4: $d_3^{k+1} = \mathcal{T}_{\lambda_2 \beta}(W u^{k+1} + b_3^k)$;

Step 5: $d_4^{k+1} = \max(b_4^k + u^{k+1}, 0)$;

Step 6: $b_1^{k+1} = b_1^k + (H u^{k+1} - d_1^{k+1})$;

Step 7: $b_2^{k+1} = b_2^k + (\nabla u^{k+1} - d_2^{k+1})$;

Step 8: $b_3^{k+1} = b_3^k + (W u^{k+1} - d_3^{k+1})$;

Step 9: $b_4^{k+1} = b_4^k + (u^{k+1} - d_4^{k+1})$.

End

end

Algorithm 2 The framelet based regularization for Poissonian image deconvolution using split Bregman iteration (PIDSB-FA).

Initialize: Choose $u^0 = g$, $d_1^0 = Hg$, $d_2^0 = Wg$, $d_3^0 = g$, and $b_1^0 = b_2^0 = b_3^0$. Set $k = 0$.

while stopping criterion is not satisfied **do**

$k = 0$ to n_{MaxIter}

Step 1: Solve $u^{k+1} = (H^T H + 1)^{-1} [H^T (d_1^k - b_1^k) + W^T (d_2^k - b_2^k)]$;

Step 2: $d_1^{k+1} = \frac{1}{2}(s^k + \sqrt{(s^k)^2 + 4\alpha g})$, where $s^k = b_1^k + H u^{k+1} - \alpha$;

Step 3: $d_2^{k+1} = \mathcal{T}_{\lambda \alpha}(W u^{k+1} + b_2^k)$;

Step 4: $d_3^{k+1} = \max(b_3^k + u^{k+1}, 0)$;

Step 5: $b_1^{k+1} = b_1^k + (H u^{k+1} - d_1^{k+1})$;

Step 6: $b_2^{k+1} = b_2^k + (W u^{k+1} - d_2^{k+1})$;

Step 7: $b_3^{k+1} = b_3^k + (u^{k+1} - d_3^{k+1})$.

End

end

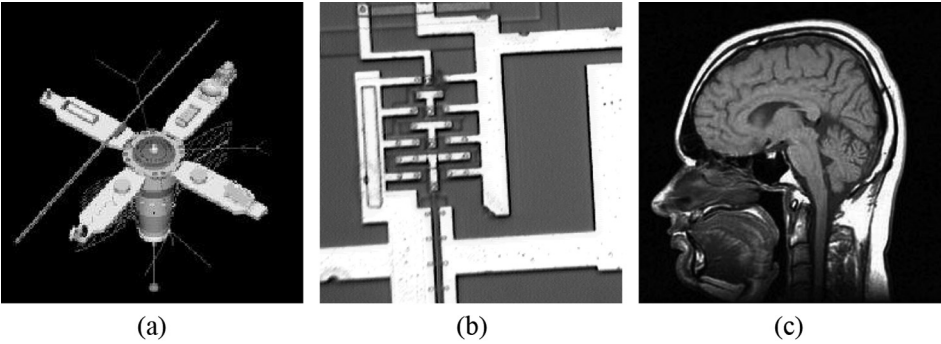


Fig. 2 Three kinds of test images of the example 1. (a) Satellite. (b) Circuit. (c) Brain.

$\|u^k - u^{k-1}\|_2 / \|u^k\|_2$, where u_k and u_{k-1} denote the image vector at the k th and $(k - 1)$ th iteration, respectively.

Motivated by Ref. 25, we observe that the stopping criterion in that paper is similar to the classic discrepancy principle²⁶ in the case of Gaussian noise. In the case of Poisson noise, the stopping criterion of both PDSB-FA and PDSB-TVFA methods is set to satisfy

$$J(u^k) = \sum_{i=1}^N \left\{ g_i \ln \frac{g_i}{(Hu^k)_i} + (Hu^k)_i - g_i \right\} \leq J_{\text{exact}},$$

where J_{exact} is the value of the function $J(u)$ for the exact original image and the maximum iteration number is set to be 300.

Table 1 PSNR (dB) of four methods (the numbers 1, 2 and 3 denote Satellite, Circuit, and Brain image, respectively).

Image	I_{max}	Type	Obs.	PIDSplit+ ⁴	PIDAL-TV ¹³	PDSB-FA	PDSB-TVFA
1	1000	Gaussian	22.59	24.44	24.52	24.59	25.19
		Defocus	22.59	25.05	25.90	25.97	26.56
		Uniform	22.92	26.16	26.65	26.68	26.98
	5000	Gaussian	22.67	25.04	25.45	25.50	25.46
		Defocus	22.69	26.76	27.42	27.55	27.45
		Uniform	23.00	27.81	28.45	28.83	28.50
	1000	Gaussian	20.88	24.58	25.51	25.78	25.58
		Defocus	21.02	25.18	25.92	26.52	25.99
		Uniform	21.09	27.63	27.87	29.23	28.54
2	5000	Gaussian	21.12	25.57	26.50	27.50	27.41
		Defocus	21.29	27.80	28.12	29.90	29.09
		Uniform	21.36	29.89	30.75	31.57	31.76
	1000	Gaussian	23.09	27.05	27.97	28.20	28.02
		Defocus	23.41	27.36	27.94	28.64	28.37
		Uniform	23.95	28.29	28.46	29.26	29.28
	5000	Gaussian	23.28	27.54	28.59	29.21	29.51
		Defocus	23.60	28.08	29.29	30.23	30.46
		Uniform	24.18	29.41	30.28	31.18	31.26

The numerical experiments have been executed in Matlab 7.10.0 on a personal computer with a dual core desktop with 2.93 GHz processor and 2 G physical memory.

The objective quality of the restored images are evaluated quantitatively by the peak signal-to-noise ratio (PSNR) which is defined as follows:

$$\text{PSNR} = 10 \log_{10} \frac{L^2}{\text{MSE}},$$

where

$$\text{MSE} = \frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} (x_{ij} - y_{ij})^2,$$

where L is the maximum possible pixel value of the image and $N_x \times N_y$ is the image size. PSNR is usually preferred for comparing images with different dynamic ranges.

For comparison, the Poissonian image deconvolution methods proposed by Setzer et al.⁴ (PIDSplit+) and Figueiredo et al.¹³ (PIDAL-TV) are also tested. The

Table 2 Parameters setting of the results of Table 1 for the proposed methods (the numbers 1, 2 and 3 denote Satellite, Circuit, and Brain image, respectively).

Image	$I_{\max}$	Type	PIDSB-FA (λ)	PIDSB-TVFA (λ_1, λ_2)
1	1000	Gaussian	0.02,	0.003, 0.01
		Defocus	0.02,	0.002, 0.01
		Uniform	0.02,	0.002, 0.005
	5000	Gaussian	0.004,	0.0002, 0.0005
		Defocus	0.004,	0.0002, 0.001
		Uniform	0.006,	0.0002, 0.001
2	1000	Gaussian	0.04,	0.004, 0.01
		Defocus	0.04,	0.004, 0.01
		Uniform	0.04,	0.004, 0.01
	5000	Gaussian	0.005,	0.0002, 0.001
		Defocus	0.005,	0.0002, 0.001
		Uniform	0.01,	0.0002, 0.003
3	1000	Gaussian	0.04,	0.004, 0.03
		Defocus	0.04,	0.004, 0.03
		Uniform	0.04,	0.004, 0.03
	5000	Gaussian	0.015,	0.015, 0.0035
		Defocus	0.015,	0.015, 0.0035
		Uniform	0.015,	0.015, 0.0035

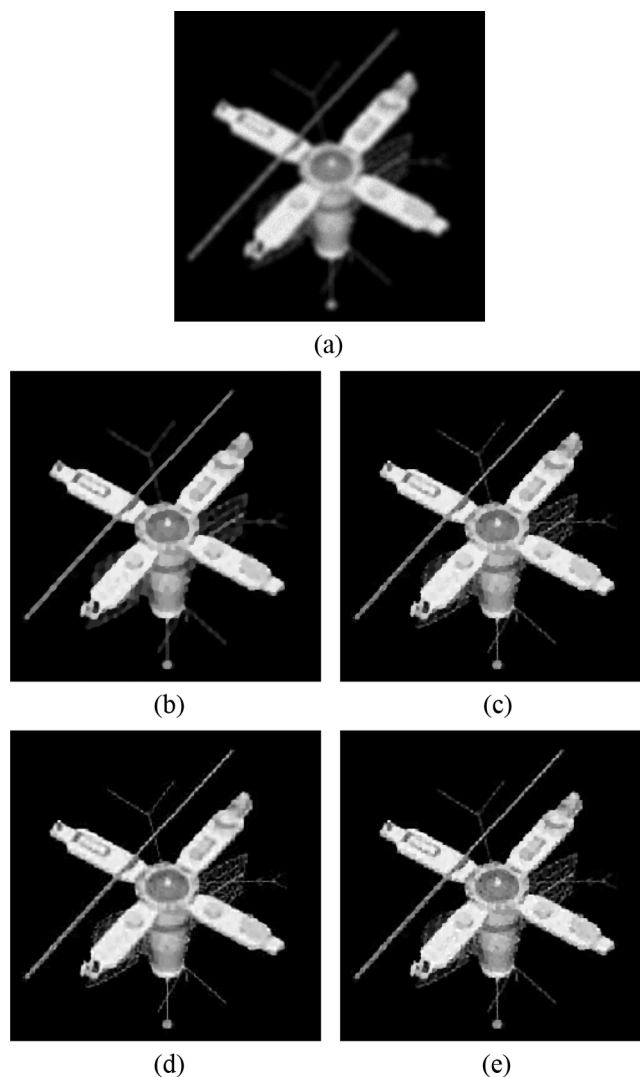


Fig. 3 Deconvolution of the Satellite image. (a) Blurred images by an out of focus kernel with radius 3 and corrupted by Poisson noise with $I_{\max}=2000$ (PSNR = 22.65 dB). (b) Restored image by the PIDSplit+ method (PSNR = 25.46 dB). (c) Restored image by the PIDAL-TV method (PSNR = 26.56 dB). (d) Restored image by the PIDSB-FA method (PSNR = 26.69 dB). (e) Restored image by the PIDSB-TVFA method (PSNR = 26.56 dB).

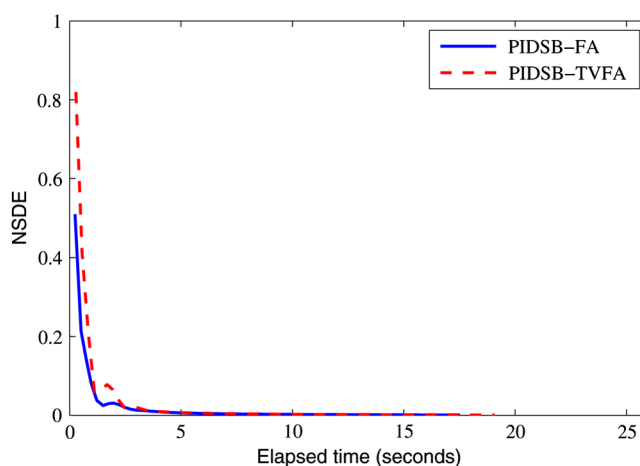


Fig. 4 The curves of the normalized step difference energy versus the elapsed time for the PIDSB-TV and PIDSB-TVFA methods.

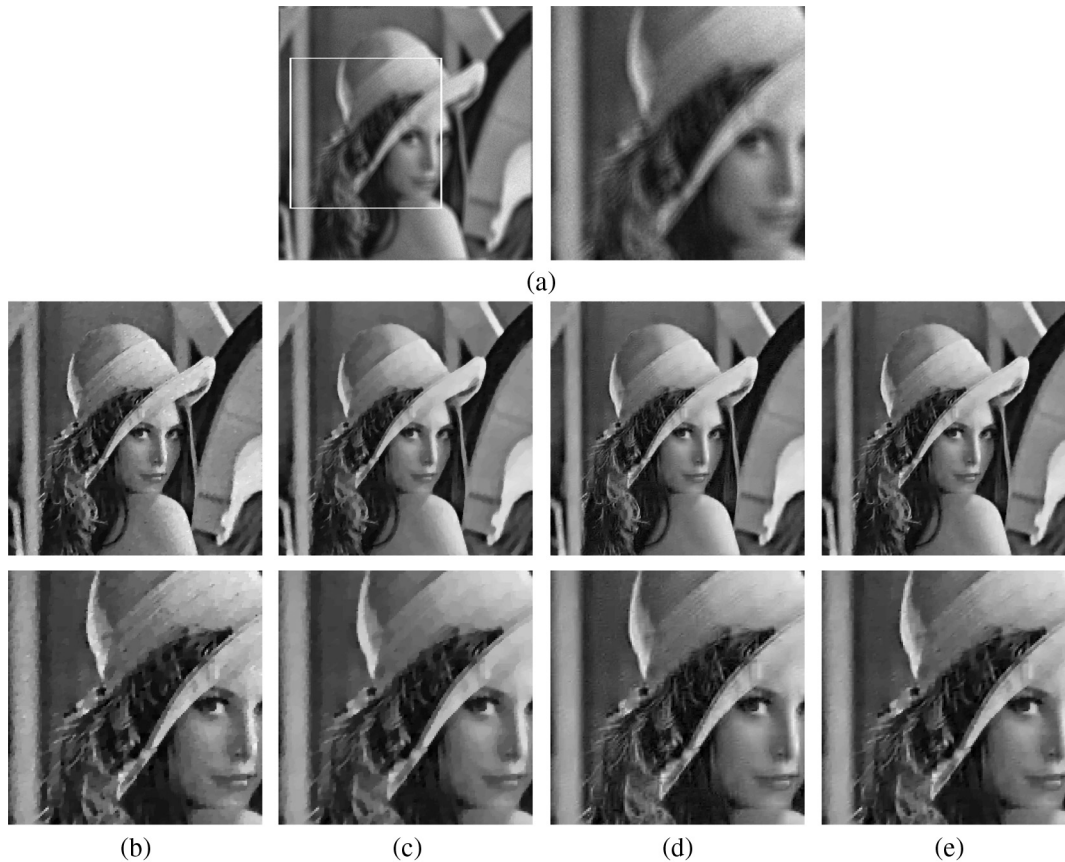


Fig. 5 Deconvolution of the Lena image. (a) Blurred by the customized kernel and then corrupted by Poisson noise (PSNR = 23.62 dB). (b) Restored image by the PIDSplit+ method (PSNR = 29.15 dB). (c) Restored image by the PIDAL-TV method (PSNR = 29.28 dB). (d) Restored image by the PIDSB-FA method (PSNR = 29.82 dB). (e) Restored image by the PIDSB-TVFA method (PSNR = 29.92 dB).

parameters used in the PIDSplit+ and PIDAL-TV methods are set according to the suggested methods from their papers and we properly adjust these parameters for obtaining the best restoration in terms of PSNR. The methods, suggested in this work, were tested on several simulated images.

The problem for Eq. (11) via split Bregman method should have four different parameters. To facilitate the expression, we use α , γ , β , μ for constraints $d_1 = Hu$, $d_2 = u$, $d_3 = Wu$, $d_4 = \nabla u$, respectively. We have to adjust these parameters repeatedly for obtaining the optimal results. However, it is very time-consuming. In our experiments, we found that we can achieve the satisfactory results by using the rule of thumb: $\alpha = \gamma = \mu$.

Example 1. To demonstrate the performance of the proposed method on a large test set, we choose the three kinds of test images which include the Satellite, Circuit and Brain images in astronomy, microscope, medicine fields, respectively. They are shown in Fig. 2. In this example, we consider two Poisson noise levels and three blur types [Gaussian blur (size: 15×15 , standard deviation: 1.8), out-of-focus blur (size: 7×7), and uniform blur (size: 5×5)]. The comparisons and PSNR values of our methods and PIDSplit+ and PIDAL-TV are shown in Table 1. It is observed that PIDSB-FA and PIDSB-TVFA produce a much higher PSNR value in every noise level and blur type compared with PIDSplit+ and PIDAL-TV. The selected parameters are presented in Table 2. The parameters were selected manually to provide

the best PSNR results. The adjustment of the parameters are dependent on the noise levels and blur intensity. It can be easily seen, from the parameters table, that we do not need to tune up the parameters rigorously for the different blur types under the same noise level for the two images and two methods due to the robustness of the algorithms. Particularly, we set $\alpha = 10/\lambda$ and $\alpha = 50/\lambda$ for Satellite and Circuit image, respectively, when we consider the PIDSB-FA method. When we perform the experiments for PIDSB-TVFA method, we take $\alpha = 5/\lambda_1$, $\beta = 50/\lambda_2$ for all noise levels and blur types. The degraded image and corresponding restored images by PIDSplit+, PIDAL-TV, PIDSB-FA, and PIDSB-TVFA methods are shown from Fig. 3(a) to 3(e), respectively. We note that both PIDSB-FA and PIDSB-TVFA methods obtain the satisfactory restored results, while the PIDSB-FA method restores more details and looks sharper as can be seen from the Fig. 3(d). It appears that Fig. 3(e) restored by the PIDSB-TVFA method is better in comparison with the Fig. 3(c) restored by the PIDAL-TV method visually although the two methods obtain the same PSNR values. In Fig. 3(b), the restored image loses some fine details. In order to present the convergence of the proposed two methods, we show the graphs of the NSDE of our methods in Fig. 4. The two curves are obtained for the Satellite image with the uniform of radius equal to five and $I_{\max}$ equals to 1000 using PIDSB-FA and PIDSB-TVFA methods, respectively. It is observed

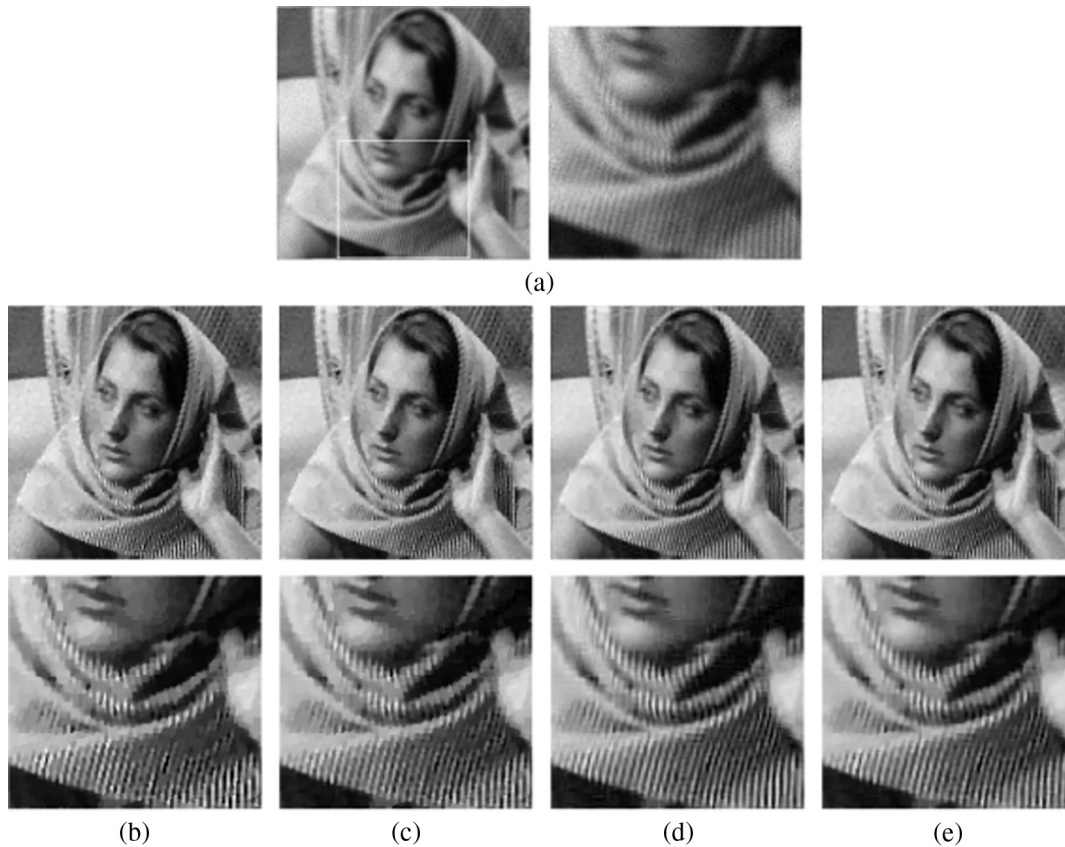


Fig. 6 Deconvolution of the Barbara image. (a) Blurred by the uniform kernel of size 5×5 and then corrupted by Poisson noise with $I_{\max}=1000$ (PSNR = 22.31 dB). (b) Restored image by the PIDSplit+ method (PSNR = 25.78 dB). (c) Restored image by the PIDAL-TV method (PSNR = 26.04 dB). (d) Restored image by the PIDSB-FA method (PSNR = 26.68 dB). (e) Restored image by the PIDSB-TVFA method (PSNR = 26.52 dB).

from Fig. 4 that the proposed methods can reach the convergence by taking less than 20 s for a 250×250 image. Thus, the convergence of the proposed methods is highlighted.

Example 2. We restore Lena image blurred by customized blur $([0\ 1\ 1\ 1\ 1\ 0\ 0; 1\ 1\ 0\ 0\ 0\ 1\ 0; 0\ 0\ 0\ 0\ 1\ 1\ 0; 0\ 0\ 1\ 1\ 0\ 0\ 0; 0\ 1\ 1\ 0\ 0\ 1\ 1; 0\ 1\ 0\ 0\ 1\ 1\ 0; 0\ 1\ 1\ 1\ 1\ 0\ 0]/22)$ and additionally contaminated by Poisson noise with $I_{\max}=2000$. The parameters used are $\lambda=0.04$ and $\alpha=40/\lambda$ for the PIDSB-FA method and $\lambda_1=0.002$, $\lambda_2=0.02$, $\alpha=5/\lambda_1$, and $\beta=50/\lambda_2$ for the PIDSB-TVFA method. The restored images, by four methods, are depicted in Fig. 5. The Lena image contains a nice mixture of detail, flat regions, shading, and texture. It can be seen from Fig. 5, that many artifacts can be detected obviously in the image restored by the PIDSplit+ method while the PIDAL-TV method removes the artifacts well, however, it also removes the details and piecewise constant effect can be still seen visually. The PIDSB-FA and PIDSB-TVFA methods remarkably improve the adverse drawback produced by the PIDSplit+ and PIDAL-TV methods. Compared with the restored results shown in Fig. 5(b) and 5(c), our results in Fig. 5(d) and 5(e) contain many more fine details such as the fringe from the hat. It is also observed, that the results from our methods (PIDSB-FA and PIDSB-TVFA) are better than the PIDSplit+ and PIDAL-TV methods in terms of PSNR values. These results show that our methods can restore the degraded image quite well in an efficient manner.

Example 3. The last example is executed on the widely used image of Barbara. It contains fine details of different textures such as scarf and clothes on her. The image is corrupted by the uniform of size 5×5 and Poisson noise of $I_{\max}=1000$ as shown in Fig. 6(a). The strong Poisson noise is obviously seen in the original degraded image [Fig. 6(a)]. The parameters used are $\lambda=0.035$ and $\alpha=30/\lambda$ for the PIDSB-FA method. The values of the parameters λ_1 , λ_2 in the PIDSB-TVFA method are set as $\lambda_1=0.002$, $\lambda_2=0.02$, and we take $\alpha=5/\lambda_1$ and $\beta=50/\lambda_2$.

We compare the results from our proposed methods PIDSB-FA and PIDSB-TVFA against those from the PIDSplit+ and PIDAL-TV methods. In order to facilitate the comparison, the close-ups of the restored images are also shown. It can be seen that the PIDSB-FA is the best performer among all four methods. Although the PIDSplit+ removes most of the noise, it still gives more heterogeneous results.

The PIDSB-FA method suppresses noise successfully while preserving significantly more details. For instance, the noise in the face and the flat area of the image in Fig. 6(d) are smoothed and the more texture details and features on the scarf in Fig. 6(d) are preserved well without being degraded by noise. The main reason may be that the linear B-spline framelet based approach adaptively applies difference operators in different regions of an image according to the singularities of the latent image to be restored.

Thus, the PDSB-FA method obtains superior results to those of the PDSplit+ and PIDAL-TV methods. The PDSB-TVFA method also obtains the pleasing restored result in suppressing noise and preserving the details in comparison to the PDSplit+ and PIDAL-TV methods. Moreover, we can also observe that the PDSB-FA and PDSB-TVFA methods yield higher PSNR values than those of the PDSplit+ and PIDAL-TV methods.

5 Conclusions

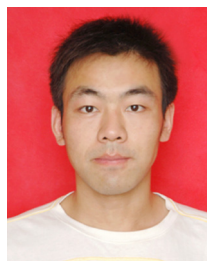
In this paper, we propose two new sparsity prior based Poissonian image deconvolution algorithms, PDSB-FA and PDSB-FATV, which are derived by the split Bregman method. The large scale problem, yielded by the proposed models, are efficiently solved using the split Bregman method. Numerical comparisons, with the TV based deconvolution algorithm (PDSplit+) and the augmented Lagrangian method with the alternating direction method of multipliers (PIDAL-TV), are given to illustrate the advantages of our algorithms. Furthermore, our algorithms produce nonnegative restored images and the convergence can be guaranteed by known convergence results of the split Bregman method.

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Houzhang Fang received his MS degree in applied mathematics from Huazhong University of Science and Technology (HUST), Wuhan, China, in 2010. Currently, he is pursuing his PhD degree in the School of Automation, HUST. His research interests include image processing and computer vision.



Luxin Yan received the BS degree in electronic communication engineering and the PhD degree from Huazhong University of Science and Technology (HUST), Wuhan, China, in 2001 and 2007, respectively. He is currently an assistant professor and director with the School of Automation, HUST. His research interests include multispectral image processing, pattern recognition, and development of embedded real-time video/image processing system.